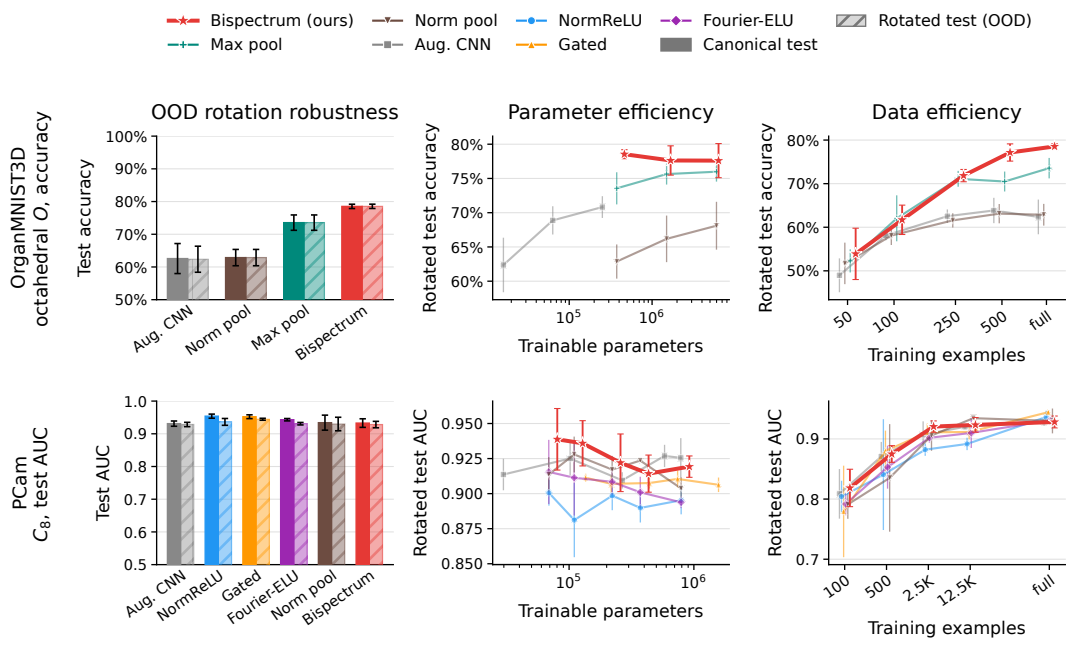

Bispectrum (ours)
Max pool
Norm pool
Aug. CNN
NormReLU
Gated
Fourier-ELU
Canonical test
Rotated test (OOD)
OrganMNIST3D
octahedral O, accuracy
OOD rotation robustness
Test accuracy
100%
90%
80%
70%
60%
50%
Aug. CNN
Norm pool
Max pool
Bispectrum
Parameter efficiency
Rotated test accuracy
80%
75%
70%
65%
60%
10^5
10^6
Trainable parameters
Data efficiency
Rotated test accuracy
80%
70%
60%
50%
50
100
250
500
full
Training examples
PCam
C_8, test AUC
Test AUC
1.0
0.9
0.8
0.7
0.6
0.5
Aug. CNN
NormReLU
Gated
Fourier-ELU
Norm pool
Bispectrum
Rotated test AUC
0.950
0.925
0.900
0.875
0.850
10^5
10^6
Trainable parameters
Rotated test AUC
0.9
0.8
0.7
100
500
2.5K
12.5K
full
Training examples